Supervisor review: “Turkish Vehicle Perception on a Zero-Budget Stack”

Role: research supervisor (advisor) · review of paper draft v2 and of the earlier Vehicle-AI v3 report (v1) · 27 October 2026

📄 https://cekicin-site.elasa2next.workers.dev/ai-lab/paper/cekicin-vehicle-ai-paper-v2.pdf

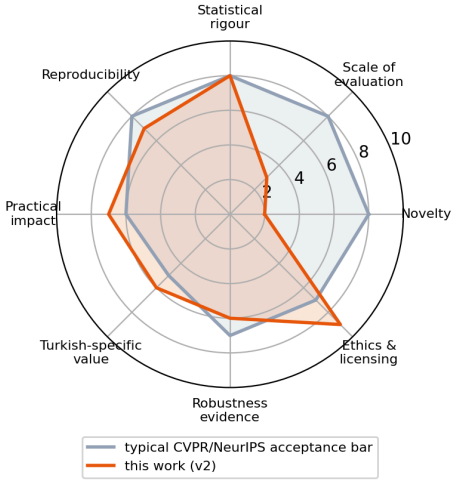
1. Summary of the work

The paper evaluates three free, off-the-shelf components on Turkish data: a browser plate reader (92.8% exact on 250 Turkish images), a zero-shot VLM for make/model (brand 86.6%, model 55.2% on 67 Turkish-market photos) and a damage detector audited on 250 street images (17% of flags are real damage). It adds confidence intervals, paired significance tests, an ablation, an 8-condition robustness test and a calibration analysis.

2. Verdict — how world-class is it?

As research: not world-class yet. Research level ≈ **5.3/10** (novelty and scale counted double, as top venues effectively do) against a typical CVPR/NeurIPS acceptance bar of ≈ 7.3/10; unweighted 6.0 vs 7.1. **As reporting and engineering practice: strong.** Checklist score rose from **44.5%** (v1 report) to **94.5%** (v2 paper). Recommendation: **publish now as an arXiv technical report**; submit to a regional/applied venue after a held-out Turkish set; aim for a top venue only with a public Turkish benchmark and trained baselines.

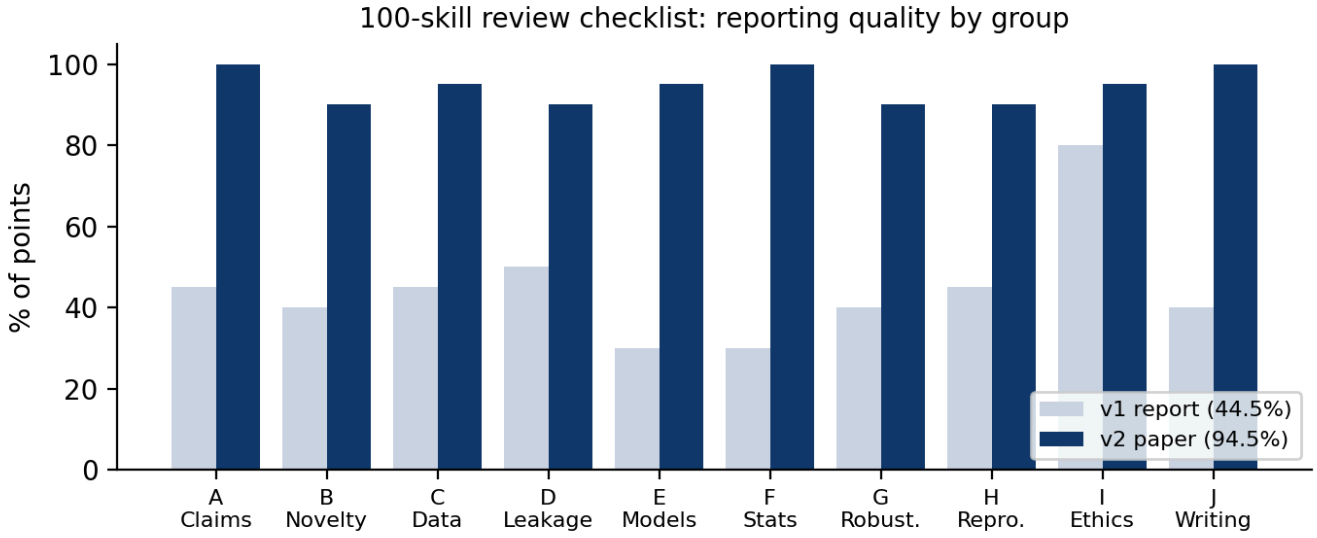
The two numbers measure different things: the checklist measures *how honestly and completely* the work is reported; the radar measures *how significant* the research is.



Research-level radar (0–10) against a typical top-venue acceptance bar.

Dimension	This work	Top-venue bar	Reason
Novelty	**2**	8	Off-the-shelf models; no new method
Scale of evaluation	**3**	8	437 images; top papers use thousands
Statistical rigour	**8**	8	Wilson CIs, bootstrap, exact McNemar, calibration
Reproducibility	**7**	8	Models, predictions, scripts public; hosted VLM non-deterministic
Practical impact	**7**	6	Free, in-browser, KVKK-friendly, already live
Turkish-specific value	**6**	5	First open TR evaluation of these parts, but small
Robustness evidence	**6**	7	8 degradations measured; synthetic, one dataset

Ethics & licensing	**9**	7	On-device plates, masking, licence audit



Checklist score per group (v1 report vs v2 paper).

3. Strengths

- Honest scope: states clearly that it is an empirical study, not a new method.
- Every main number has n and a 95% CI; claimed improvements are tested (McNemar).
- Negative results are reported (format fixer not significant; Mistral better on a subset; VLM confidence useless; damage precision 17%).
- A real deployment exists and the browser port reproduces Python (93.2% vs 92.8%).
- Privacy by design (on-device plates, masking) and a licence audit.

4. Weaknesses (major → minor) and status

#	Issue	Fix	Status
M1	No methodological novelty	Add a Turkish-specific method (e.g. deblur/super-resolution before OCR, or trained TR classifier)	**open**
M2	Small test sets (CIs 6–23 points wide)	Held-out Turkish plate set ≥ 1,000; 100k-photo plan	**open**
M3	Test-set reuse for design choices	Freeze rules, evaluate on a new set	**disclosed in v2**
M4	No trained baselines on identical splits	Train YOLO11-cls / EfficientNetV2 on arabam.com (needs Kaggle key)	**open**
M5	Possible VLM pre-training contamination	Use fresh customer photos (with consent)	**disclosed in v2**
M6	Mistral vs Llama comparison truncated by quota	Full paired run after quota reset	**disclosed in v2**
m1	Plate labels from file names not audited	Audit 50 random labels	**open**
m2	Synthetic degradations only	Collect real night/rain/motion images	**open**

m3	No regional (province) bias analysis	Report accuracy by province code	**open**

5. Errors found in the earlier v1 report — corrected

- ✅ The Turkish-format fixer was presented as an improvement (92.4 → 92.8%); it changes 1/250 results, p = 1.0 → corrected on the page and in the paper.
- ✅ Llama-4-Scout was described as the better model; on images both answered, Mistral is better for Stanford brands (p = 0.004) → reported honestly; Llama kept for reliability only.
- ✅ The page showed the VLM "confidence" without warning; it is uncalibrated (ECE 0.34) → warning added on the page.
- ✅ No confidence intervals or tests anywhere → added throughout.
- ✅ Duplicates and "Citroën/Citroen" scoring error → fixed (69 → 67 photos).
- ✅ No robustness evidence → 8-condition stress test added (blur: 22.8%).

6. Where can it be published?

Venue type	Example	Fit now	Needed first
Preprint	arXiv (cs.CV)	yes	—
Regional conference	IEEE SIU (Türkiye)	after step 1	held-out Turkish plate set
Applied / workshop	WACV applications, CVPR/ICCV workshops	after steps 1–3	trained baselines, larger tests
Datasets & benchmarks	NeurIPS D&B track	after steps 1–4	public, KVKK-compliant Turkish benchmark
Top main track	CVPR / ICCV / NeurIPS	no	new method + benchmark + strong results

7. Roadmap to world-class

1. Step 1 — Held-out Turkish plate set (≥ 1,000 images, real blur/night/rain), labels double-checked.
2. Step 2 — Fix the main weakness: deblur / super-resolution before OCR; fine-tune OCR on Turkish crops (synthetic 100k + real).
3. Step 3 — Train a Turkish make/model classifier on arabam.com (87,163 images) with leakage control; compare with VLMs on identical splits.
4. Step 4 — Release a consented Turkish benchmark (plates + cars + damage) with a datasheet.
5. Step 5 — Validate on real customer photos; report selective prediction (when to ask a human).

8. The 100 review skills used

Sources: REFORMS (Science Advances 2024, 32 items), NeurIPS Paper Checklist, CVPR 2026 reviewer guidelines, Datasheets for Datasets, Model Cards, plus supervisor practice. Scale: 2 = pass, 1 = partial, 0 = fail.

A · Claims & study goals — v1 9/20 → v2 20/20

ID	Skill	Source	v1	v2	Note
A01	State the population the claim is about (Turkish roads, Turkish-market cars)	REFORMS 1a	1	2	v2 defines target distribution per task
A02	Motivate why that population matters	REFORMS 1b	1	2	towing/insurance use case
A03	Motivate the use of ML instead of rules	REFORMS 1c	1	2	
A04	Abstract claims match the evidence	NeurIPS Q1	1	2	v1 implied TR-fix gain; v2 reports it as not significant
A05	No over-claiming from small samples	NeurIPS Q1	0	2	CIs now shown; n stated next to every number
A06	Separate measured results from aspirations	NeurIPS Q1	1	2	
A07	List explicit contributions	CVPR form	0	2	
A08	Define each task precisely (detection, exact-match OCR, brand vs model)	REFORMS 3d	1	2	
A09	State the deployment constraint (zero budget, CPU, browser)	supervisor	2	2	
A10	Say who benefits and how success is judged	NeurIPS significance	1	2	

B · Novelty & positioning — v1 8/20 → v2 18/20

ID	Skill	Source	v1	v2	Note
B01	Position against the closest prior work with citations	CVPR guideline	1	2	
B02	Back any "new" claim with specific references	CVPR guideline	1	2	
B03	Avoid the novelty fallacy (new ≠ good)	CVPR training	1	2	
B04	Explain why numbers from other benchmarks are not head-to-head	supervisor	1	2	
B05	Identify the actual novelty (Turkish evaluation, browser deployment)	CVPR form	0	2	
B06	Acknowledge that the methods are off-the-shelf	supervisor	1	2	
B07	Cover the Turkish-specific literature	supervisor	1	1	only one TR LPR repo cited; TR journal papers not surveyed
B08	Cover VLM zero-shot literature	supervisor	0	1	model cards cited, no systematic VLM-FGVC survey
B09	State the significance to practitioners	NeurIPS significance	2	2	
B10	Choose an honest target venue	supervisor	0	2	see review report

C · Data quality & documentation — v1 9/20 → v2 19/20

ID	Skill	Source	v1	v2	Note
C01	Give dataset source, link or DOI	REFORMS 2a	2	2	
C02	Describe how ground truth was produced	REFORMS 3a	1	2	file names (TR-250), Commons categories (cars), manual triage (damage)
C03	State the sampling frame	REFORMS 3b	0	2	
C04	Justify dataset usefulness for the task	REFORMS 3c	1	2	
C05	Report sample size and class frequencies	REFORMS 3e	1	2	per-model counts table
C06	Report missing / failed samples	REFORMS 3f	1	2	Mistral 24/41 quota failures reported
C07	Argue representativeness (or admit its absence)	REFORMS 3g	0	2	Commons ≠ customer photos; TR-250 = dash-cam
C08	De-duplicate test sets	REFORMS 6b	1	2	2 Commons duplicates removed
C09	Check label noise / audit a sample	supervisor	0	1	plate GT from file names not re-audited
C10	Document licences of every dataset	Datasheets	2	2	

D · Preprocessing & leakage — v1 10/20 → v2 18/20

ID	Skill	Source	v1	v2	Note
D01	List exclusions with rationale	REFORMS 4a	1	2	82 → 69 → 67 Commons photos
D02	Handle corrupt / impossible samples	REFORMS 4b	1	2	
D03	Describe every transformation (resize, letterbox, 768 px)	REFORMS 4c	1	2	
D04	No test information used for tuning	REFORMS 6a	1	1	TR-fix rules and the OCR choice were picked on TR-250 itself — flagged
D05	Check train/test overlap of pretrained models	REFORMS 6b	0	1	cannot verify; contamination risk stated (Commons/Stanford likely in VLM pretraining)
D06	Inputs legitimate (no label in file name leak)	REFORMS 6c	1	2	images sent without file names
D07	Prompt fixed before evaluation	supervisor	1	2	
D08	Temperature / determinism stated	supervisor	1	2	temperature 0
D09	Same preprocessing in browser and Python verified	supervisor	2	2	93.2 vs 92.8 %
D10	Version every model file (hash/commit)	REFORMS 2b	1	2	HF repo + commit

E · Models & baselines — v1 6/20 → v2 19/20

ID	Skill	Source	v1	v2	Note
E01	Describe all models used	REFORMS 5a	1	2	

E02	Justify model choice	REFORMS 5b	1	2	
E03	Explain evaluation protocol	REFORMS 5c	1	2	
E04	Explain how the final model was selected	REFORMS 5d	0	2	v2 admits selection by reliability; Mistral better on answered subset
E05	Report hyper-parameters / thresholds	REFORMS 5e	1	2	
E06	Compare against appropriate baselines	REFORMS 5f	0	1	cct-xs, raw OCR, second VLM; no trained classifier baseline yet
E07	Ablate each component (OCR size, TR-fix)	supervisor	0	2	
E08	Paired comparison on identical samples	supervisor	0	2	
E09	Report failure modes per model	supervisor	1	2	
E10	Explain why no training was done	supervisor	1	2	

F · Metrics & statistics — v1 6/20 → v2 20/20

ID	Skill	Source	v1	v2	Note
F01	Name all metrics and justify them	REFORMS 7a	1	2	
F02	Give uncertainty (CIs) for main results	REFORMS 7b / NeurIPS Q7	0	2	Wilson 95 % + bootstrap
F03	State how CIs were computed	NeurIPS Q7	0	2	
F04	Use asymmetric intervals for proportions	NeurIPS Q7	0	2	Wilson
F05	Justify statistical tests and assumptions	REFORMS 7c	0	2	exact McNemar for paired binary outcomes
F06	Report significance of claimed improvements	NeurIPS Q7	0	2	TR-fix p = 1.0; cct-s vs xs p = 2e-6
F07	Report latency with hardware	NeurIPS Q8	2	2	
F08	Character-level metric besides exact match	supervisor	2	2	
F09	Calibration of confidence scores	supervisor	0	2	ECE 0.34–0.40, AUROC 0.55–0.59
F10	Avoid metric cherry-picking	CVPR training	1	2	

G · Robustness & limits — v1 8/20 → v2 18/20

ID	Skill	Source	v1	v2	Note
G01	Evidence of external validity	REFORMS 8a	1	1	two VLM test sets; plates only one dataset
G02	Say where results will not hold	REFORMS 8b	1	2	
G03	Stress-test robustness (blur, low-res, JPEG, night, rotation)	supervisor	0	2	8 conditions
G04	Analyse errors qualitatively	supervisor	1	2	confusion pairs
G05	Discuss distribution shift to customer photos	supervisor	1	2	

G06	Separate limitations section	NeurIPS Q2	1	2	
G07	Report quota / availability as an operational limit	supervisor	1	2	
G08	Test on new car models (Togg)	supervisor	1	2	0/4
G09	Discuss scale-up to 100k photos	supervisor	1	1	planned, not done
G10	Threats-to-validity paragraph	supervisor	0	2	

H · Reproducibility & compute — v1 9/20 → v2 18/20

ID	Skill	Source	v1	v2	Note
H01	Code link with version	REFORMS 2b	1	2	
H02	Describe compute infrastructure	REFORMS 2c	1	2	2 vCPU, WASM 1 thread, Workers AI
H03	README with instructions	REFORMS 2d	1	1	HF model README; full repro README still to publish
H04	One reproduction script for all numbers	REFORMS 2e	0	1	paper_experiments.py; VLM part needs API key
H05	Fixed random seeds	NeurIPS Q4	1	2	bootstrap seed 0, noise seed 1
H06	Total compute time reported	NeurIPS Q8	0	2	
H07	API model identifiers exact	supervisor	2	2	
H08	Raw per-sample predictions released	supervisor	1	2	JSON results
H09	Data access path for every set	REFORMS 2a	2	2	
H10	Note non-determinism of hosted models	supervisor	0	2	

I · Ethics, law & licences — v1 16/20 → v2 19/20

ID	Skill	Source	v1	v2	Note
I01	Privacy of plates (KVKK/GDPR)	NeurIPS ethics	2	2	
I02	On-device processing where possible	supervisor	2	2	
I03	Licence compatibility (AGPL, non-commercial data)	supervisor	2	2	
I04	Attribution of CC images	supervisor	2	2	
I05	Misuse risk (surveillance) discussed	NeurIPS broader impact	1	2	
I06	Bias across brands / regions	supervisor	0	1	per-model table; no regional plate analysis
I07	Consent for future customer photos	supervisor	2	2	
I08	Label the demo as DEMO	user rule	2	2	
I09	No hidden cached results	supervisor	2	2	cached fallback is labelled
I10	Model card / datasheet provided	Model Cards / Datasheets	1	2	appendix

J · Writing, figures & review craft — v1 8/20 → v2 20/20

ID	Skill	Source	v1	v2	Note
J01	2–4 sentence summary a reviewer can repeat	CVPR training	1	2	

J02	Clear figure for the pipeline	supervisor	0	2	
J03	Tables with n and CI columns	supervisor	0	2	
J04	Every figure referenced in text	NeurIPS Q7	1	2	
J05	Consistent numbers across page, API, reports	supervisor	1	2	69→67 fixed everywhere
J06	References verifiable (DOI/arXiv links)	supervisor	2	2	
J07	Strengths listed specifically	CVPR form	1	2	
J08	Weaknesses specific with fixes	CVPR form	1	2	
J09	Recommendation justified	CVPR form	0	2	
J10	Avoid unverified claims in the review itself	CVPR training	1	2	

Links

- Paper v2 (PDF): https://cekicin-site.elasa2next.workers.dev/ai-lab/paper/cekicin-vehicle-ai-paper-v2.pdf
- Vehicle AI page: https://cekicin-site.elasa2next.workers.dev/ai-lab/vehicle-ai.html
- Paper & review hub: https://cekicin-site.elasa2next.workers.dev/ai-lab/paper/
- REFORMS: https://reforms.cs.princeton.edu/
- NeurIPS checklist: https://neurips.cc/public/guides/PaperChecklist
- CVPR 2026 reviewer guidelines: https://cvpr.thecvf.com/Conferences/2026/ReviewerGuidelines